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JEE MAIN PHYSICS DPP: RELATIVE VELOCITY IN 1D

Topic: Relative Velocity in One Dimension – Concept, Collision & Overtake Problems | **Time: 45 Min** | **Max Marks: 80**

Marking Scheme: **+4, -1** | Designed & Curated by: **Syed Mazhar Ali**

Q1. Two trains A and B of lengths 150 m and 250 m are moving in opposite directions on parallel tracks at speeds 72 km/h and 36 km/h respectively. The time taken by the two trains to completely cross each other is:

(a) 20.0 s
(b) 13.33 s
(c) 40.0 s
(d) 10.0 s

Q2. A police van moving on a highway at 30 km/h fires a bullet at a thief's car speeding away in the same direction at 192 km/h. If the muzzle speed of the bullet is 150 m/s, the speed with which the bullet hits the thief's car is:

(a) 150 m/s
(b) 95 m/s
(c) 105 m/s
(d) 140 m/s

Q3. A person walks up a stationary escalator in time t_1. When standing still on a moving escalator, the escalator takes him up in time t_2. The time taken by the person to walk up the moving escalator is:

(a) $\dfrac{t_1+t_2}{2}$
(b) $\sqrt{t_1 t_2}$
(c) $\dfrac{t_1-t_2}{2}$
(d) $\dfrac{t_1 t_2}{t_1+t_2}$

Q4. Two cars A and B are travelling in the same direction with velocities $v_A = 20$ m/s and $v_B = 10$ m/s. When car A is at a distance $d = 50$ m behind B, the driver of A applies brakes producing a constant retardation a. The minimum value of a to prevent a collision is:

(a) 1.0 m/s^2
(b) 2.0 m/s^2
(c) 0.5 m/s^2
(d) 1.5 m/s^2

Q5. Two particles A and B are projected simultaneously along the same vertical line. A is dropped from a tower of height $H = 200$ m, while B is thrown upwards from the ground with velocity $u = 50$ m/s. The acceleration of A relative to B during flight is:

(a) $2g$ downwards
(b) 0
(c) g downwards
(d) g upwards

Q6. In the previous question, the time after which the two particles collide is:

(a) 2 s
(b) 5 s
(c) 4 s
(d) 10 s

Q7. A passenger in a train moving at speed v throws a ball vertically upwards with speed u inside the compartment. The trajectory of the ball as observed by another passenger seated in the same compartment is:

(a) A parabola opening downwards
(b) A straight line inclined at $\tan^{-1}(u/v)$
(c) A horizontal line
(d) A vertical straight line

Q8. Two trains, each of length 100 m, are running on parallel tracks. One overtakes the other in 20 s and passes the other in 10 s when running in opposite directions. The speeds of the two trains are:

(a) 15 m/s and 5 m/s
(b) 20 m/s and 10 m/s
(c) 12 m/s and 8 m/s
(d) 18 m/s and 6 m/s

Q9. On a two-lane road, car A is travelling with a speed of 36 km/h. Two cars B and C approach car A in opposite directions with speed of 54 km/h each. At a certain instant, when the distance $AB = AC = 1$ km, car B decides to overtake A before C does. The minimum acceleration of car B required to avoid an accident is:

(a) 0.5 m/s^2
(b) 1.0 m/s^2
(c) 1.5 m/s^2
(d) 2.0 m/s^2

Q10. A bus starts from rest with an acceleration of 1 m/s^2. A man who is 48 m behind the bus with a constant speed v runs after the bus. The minimum value of v so that the man can catch the bus is:

(a) 8 m/s

(b) 12 m/s
(c) 9.8 m/s (or $\sqrt{96} \approx 9.8$ m/s)
(d) 16 m/s

Q11. A train of length l is moving with constant acceleration a. A person at the front end tosses a coin backwards along the corridor with velocity u_0 relative to the train. The acceleration of the coin relative to the train's floor is:

(a) 0
(b) $+a$ forward
(c) $g - a$
(d) a backward (pseudo acceleration)

Q12. Two cars A and B are at positions $x_A = 0$ and $x_B = 100$ m at $t = 0$. Car A moves towards $+x$ with velocity $v_A = 30$ m/s and car B moves towards $-x$ with velocity $v_B = 20$ m/s. The position where they cross each other is:

(a) $x = 60$ m
(b) $x = 40$ m
(c) $x = 50$ m
(d) $x = 75$ m

Q13. A ball is dropped from an elevator at height h from the elevator floor. If the elevator is moving upwards with constant acceleration a_0, the time taken by the ball to reach the floor is:

(a) $\sqrt{\dfrac{2h}{g}}$
(b) $\sqrt{\dfrac{2h}{g + a_0}}$
(c) $\sqrt{\dfrac{2h}{g - a_0}}$
(d) $\sqrt{\dfrac{2h}{a_0}}$

Q14. An open freight train car is moving horizontally with constant speed v. Rain is falling vertically downwards at speed u with respect to the ground. The angle with the vertical at which rain appears to fall for an observer on the train is:

(a) $\tan^{-1}(u/v)$
(b) $\sin^{-1}(v/u)$
(c) $\tan^{-1}(v/u)$
(d) $\cos^{-1}(v/u)$

Q15. Two bodies A and B are separated by 150 m on a straight track. A moves towards B from rest with constant acceleration 2 m/s^2, while B moves towards A from rest with constant acceleration 1 m/s^2. The time when they meet is:

(a) 15 s
(b) 5 s
(c) 20 s
(d) 10 s

Q16. A passenger train is moving at 30 m/s on a track. The driver suddenly spots a freight train 100 m ahead moving on the same track in the same direction at 10 m/s. The minimum deceleration the passenger train must apply to avoid a collision is:

(a) 2.0 m/s^2
(b) 1.0 m/s^2
(c) 4.0 m/s^2
(d) 0.5 m/s^2

Q17. If body A is moving with velocity $\vec{v}_A$ and body B with velocity $\vec{v}_B$, the velocity of A relative to B ($\vec{v}_{AB}$) and velocity of B relative to A ($\vec{v}_{BA}$) satisfy:

(a) $\vec{v}_{AB} = \vec{v}_{BA}$
(b) $\vec{v}_{AB} = -\vec{v}_{BA}$
(c) $\vec{v}_{AB} \cdot \vec{v}_{BA} > 0$
(d) $\vec{v}_{AB} + \vec{v}_{BA} = 2\vec{v}_A$

Q18. A river flows from west to east at speed $v_r = 3$ km/h. A boat capable of speed $v_b = 5$ km/h in still water is steered due downstream along the 1D line of the river. The time taken to travel 16 km downstream and return to the starting point upstream is:

(a) 6 hours
(b) 8 hours
(c) 10 hours
(d) 4 hours

Q19. A particle P is at $x = 0$ moving with constant speed 12 m/s. Another particle Q starts from $x = 0$ at $t = 0$ with zero initial velocity and constant acceleration 3 m/s^2. The distance between P and Q when their relative velocity becomes zero is:

(a) 48 m
(b) 12 m
(c) 36 m
(d) 24 m

Q20. Two cars A and B travel on parallel lanes. The x-t graphs of A and B are straight lines of slopes $\tan(45°)$ and $\tan(30°)$ respectively. The velocity of A relative to B is:

(a) $\left(1 - \dfrac{1}{\sqrt{3}}\right)$ m/s
(b) $\left(1 + \dfrac{1}{\sqrt{3}}\right)$ m/s
(c) $\sqrt{3} - 1$ m/s
(d) $\dfrac{1}{\sqrt{3}}$ m/s

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	**6**	(c)	**11**	(d)	**16**	(a)
2	(c)	**7**	(d)	**12**	(a)	**17**	(b)
3	(d)	**8**	(a)	**13**	(b)	**18**	(c)
4	(a)	**9**	(b)	**14**	(c)	**19**	(d)
5	(b)	**10**	(c)	**15**	(d)	**20**	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

- $v_A = 72$ km/h $= 72 \times \frac{5}{18} = 20$ m/s.
- $v_B = 36$ km/h $= 36 \times \frac{5}{18} = 10$ m/s (opposite direction).

Relative velocity:

$$v_{\text{rel}} = v_A - (-v_B) = 20 + 10 = 30 \text{ m/s}$$

Total length to clear: $L = 150 + 250 = 400$ m.
Time:

$$t = \frac{L}{v_{\text{rel}}} = \frac{400}{30} = \frac{40}{3} \text{ s} \approx 13.33 \text{ s}$$

Sol 2. Correct Option: (c)

- Speed of police van: $v_p = 30$ km/h $= \frac{30 \times 5}{18} = \frac{25}{3}$ m/s.
- Speed of bullet relative to ground:

$$v_{bg} = v_{\text{muzzle}} + v_p = 150 + \frac{25}{3} = \frac{475}{3} \text{ m/s}$$

- Speed of thief's car: $v_t = 192$ km/h $= \frac{192 \times 5}{18} = \frac{160}{3}$ m/s.

Hitting speed (bullet relative to thief):

$$v_{\text{rel}} = v_{bg} - v_t = \frac{475}{3} - \frac{160}{3} = \frac{315}{3} = 105 \text{ m/s}$$

Sol 3. Correct Option: (d)

Let escalator length be L.

- Walking speed: $v_m = \frac{L}{t_1}$.
- Escalator speed: $v_e = \frac{L}{t_2}$.

When walking on moving escalator:

$$v_{\text{net}} = v_m + v_e = \frac{L}{t_1} + \frac{L}{t_2} = L\left(\frac{t_1 + t_2}{t_1 t_2}\right)$$

Time taken:

$$t = \frac{L}{v_{\text{net}}} = \frac{t_1 t_2}{t_1 + t_2}$$

Sol 4. Correct Option: (a)

In relative frame of car B:
$u_{\text{rel}} = v_A - v_B = 20 - 10 = 10$ m/s.
$v_{\text{rel}} = 0$ (at closest approach / just preventing collision).
$s_{\text{rel}} = 50$ m.
Using $v_{\text{rel}}^2 = u_{\text{rel}}^2 - 2a_{\text{rel}}s_{\text{rel}}$:

$$0 = (10)^2 - 2a(50)$$

$$100a = 100 \implies a = 1.0 \text{ m/s}^2$$

Sol 5. Correct Option: (b)

Both particles are in free fall under gravity, so their individual accelerations are $\vec{a}_A = -g\hat{j}$ and $\vec{a}_B = -g\hat{j}$.
Relative acceleration:

$$\vec{a}_{\text{rel}} = \vec{a}_A - \vec{a}_B = (-g) - (-g) = 0$$

Sol 6. Correct Option: (c)

Relative initial velocity:

$$u_{\text{rel}} = u_A - u_B = 0 - (-50) = 50 \text{ m/s}$$

Since relative acceleration is zero:

$$t = \frac{S_{\text{rel}}}{u_{\text{rel}}} = \frac{200 \text{ m}}{50 \text{ m/s}} = 4 \text{ s}$$

Sol 7. Correct Option: (d)

In the train's reference frame (moving at constant horizontal velocity), the passenger and the ball share the same horizontal speed. No pseudo-force exists since acceleration is zero. Hence, the ball goes straight up and down vertically.

Sol 8. Correct Option: (a)

Let speeds be v_1 and v_2 ($v_1 > v_2$). Total distance $= 100 + 100 = 200$ m.

- Same direction:

$$\frac{200}{v_1 - v_2} = 20 \implies v_1 - v_2 = 10$$

- Opposite direction:

$$\frac{200}{v_1 + v_2} = 10 \implies v_1 + v_2 = 20$$

Solving gives $v_1 = 15$ m/s and $v_2 = 5$ m/s.

Sol 9. Correct Option: (b)
- $v_A = 36$ km/h $= 10$ m/s
- $v_B = 54$ km/h $= 15$ m/s (same direction as A)
- $v_C = 54$ km/h $= -15$ m/s (opposite direction)

Time for C to reach A:

$$t = \frac{d}{v_{CA}} = \frac{1000}{15+10} = \frac{1000}{25} = 40 \text{ s}$$

In 40 s, B must cover relative distance $s_{\text{rel}} = 1000$ m relative to A:
$u_{\text{rel}} = v_B - v_A = 15 - 10 = 5$ m/s.

$$s_{\text{rel}} = u_{\text{rel}}t + \frac{1}{2}at^2 \implies 1000 = 5(40) + \frac{1}{2}a(40)^2$$
$$1000 = 200 + 800a$$
$$800a = 800 \implies a = 1.0 \text{ m/s}^2$$

Sol 10. Correct Option: (c)
In relative frame of the bus:
$u_{\text{rel}} = v$, $a_{\text{rel}} = -a = -1$ m/s^2, $s_{\text{rel}} = 48$ m.
For the man to catch the bus:

$$v_{\text{rel}}^2 \geq 0 \implies u_{\text{rel}}^2 - 2a_{\text{rel}}s \geq 0$$
$$v^2 - 2(1)(48) \geq 0 \implies v^2 \geq 96$$
$$v_{\min} = \sqrt{96} \approx 9.8 \text{ m/s}$$

Sol 11. Correct Option: (d)
In the non-inertial accelerating reference frame of the train (a forward), a fictitious pseudo-acceleration $\vec{a}_{\text{pseudo}} = -a$ (backward) acts on every free body.

Sol 12. Correct Option: (a)
Relative speed: $v_{\text{rel}} = v_A + v_B = 30 + 20 = 50$ m/s.
Time to meet:

$$t = \frac{d}{v_{\text{rel}}} = \frac{100}{50} = 2 \text{ s}$$

Position: $x = v_At = 30(2) = 60$ m.

Sol 13. Correct Option: (b)
In the elevator's accelerated frame:
Effective downward acceleration:

$$g_{\text{eff}} = g - (-a_0) = g + a_0$$

Time to hit floor:

$$h = \frac{1}{2}g_{\text{eff}}t^2 \implies t = \sqrt{\frac{2h}{g+a_0}}$$

Sol 14. Correct Option: (c)
$\vec{v}_{\text{rain}} = -u\hat{j}$, $\vec{v}_{\text{train}} = v\hat{i}$.
Relative velocity of rain:

$$\vec{v}_{\text{rel}} = \vec{v}_{\text{rain}} - \vec{v}_{\text{train}} = -v\hat{i} - u\hat{j}$$

Angle with vertical:

$$\tan\theta = \frac{v}{u} \implies \theta = \tan^{-1}\left(\frac{v}{u}\right)$$

Sol 15. Correct Option: (d)
Relative acceleration towards each other:

$$a_{\text{rel}} = a_A + a_B = 2 + 1 = 3 \text{ m/s}^2$$

Initial relative velocity $u_{\text{rel}} = 0$.

$$150 = \frac{1}{2}(3)t^2 \implies t^2 = 100 \implies t = 10 \text{ s}$$

Sol 16. Correct Option: (a)
In the frame of freight train:
$u_{\text{rel}} = 30 - 10 = 20$ m/s.
$v_{\text{rel}} = 0$ (at closest distance), $s = 100$ m.

$$0 = u_{\text{rel}}^2 - 2a_{\text{rel}}s \implies 0 = 20^2 - 2a(100)$$
$$200a = 400 \implies a = 2.0 \text{ m/s}^2$$

Sol 17. Correct Option: (b)
$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = -(\vec{v}_B - \vec{v}_A) = -\vec{v}_{BA}$.

Sol 18. Correct Option: (c)
- Downstream speed: $v_{\text{down}} = v_b + v_r = 5 + 3 = 8$ km/h.
- Upstream speed: $v_{\text{up}} = v_b - v_r = 5 - 3 = 2$ km/h.

Total time:

$$T = \frac{16}{8} + \frac{16}{2} = 2 + 8 = 10 \text{ hours}$$

Sol 19. Correct Option: (d)
Velocity of Q: $v_Q = at = 3t$.
Relative velocity is zero when $v_Q = v_P$:

$$3t = 12 \implies t = 4 \text{ s}$$

At $t = 4$ s:
$x_P = 12(4) = 48$ m.
$x_Q = \frac{1}{2}(3)(4)^2 = 24$ m.
Separation $= x_P - x_Q = 48 - 24 = 24$ m.

Sol 20. Correct Option: (a)
- $v_A = \tan(45°) = 1$ m/s
- $v_B = \tan(30°) = \frac{1}{\sqrt{3}}$ m/s

Relative velocity:

$$v_{AB} = v_A - v_B = 1 - \frac{1}{\sqrt{3}} \text{ m/s}$$